

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Basic Electronics Course Code: EC-111

Examination: B.Tech Scheme: CBS Branch - CSE (AI&ML) Semester: 2nd

Question Paper for 2nd Periodical Examinations (April 2024)

Time: Two hours

Max. Marks: 50

Note: All questions are compulsory. Answer according to knowledge levels.

Course Outcomes: At the end of this course students will be able to:

CO112.1	K2	Understand the applications of diode in rectifiers, filter circuits and wave shaping.
CO112.2	K2	Understand the difference between Bipolar Junction Transistors and Field Effect Transistors and differentiate on basis of their characteristics.
CO112.3	K4	Design simple circuits like amplifiers (inverting and non-inverting), comparators, adders, integrator and differentiator using OPAMPS
CO112.4	K4	Compile the different building blocks in digital electronics using logic gates and implement simple logic function using basic universal gates

Knowledge levels :: K1=Remember, K2=Understand, K3=Apply, K4=Analyze, K5=Evaluate, K6=Create.

PART - A

Q.	(CO)	Question	K	Marks
Q1.	(CO112.2)	Explain the operation of Enhancement type MOSFET and draw its drain and transfer characteristics?	K2	(10)
Q2.	(CO112.4)	Convert to hexadecimal and binary: $(20751.457)_8$ Simplify the following Boolean expression: $Y = (\bar{A} + B)(A + B)$	K2	(10)
Q3.	(CO112.2)	What are the Barkhausen criterion for oscillations? Draw and explain the basic circuit diagram of an RC Phase shift oscillator?	K1	(10)
Q4.	(CO112.3)	With the proper circuit diagrams, explain how Op-Amps can be used as Inverting, non-inverting Amplifiers and an Integrator. Derive the output voltage equation for all the cases.	K1	(10)

PART - B

Q.5	(CO)	Answer in brief according to marks assigned.	K	(10)
a		Why FET is called a unipolar device?		(2)
b		What is difference between Enhancement type and Depletion type MOSFET?		(2)
c		Find the resonance frequency of Wein bridge oscillator if $R = 51 \text{ Kohms}$ and $C = 0.001 \text{ microFarads}$.		(2)
d		Perform $11100.101 - 101.01$ using 2's compliment?		(2)
e		Find the decimal equivalent of binary number 11010?		(2)

(Course Advisor: Ankush Kapoor)

Vision: To achieve academic excellence and become pace setter by opting quality processes interwoven with freedom of flexibility and to mould professionals of superior quality dedicated to the progress and development of humanity.